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A Qos-Aware, Hybrid Particle Swarm Optimization-Cuckoo Search Clustering Based Multipath Routing in Wireless Sensor Networks

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Abstract
Wireless sensor networks applications have been used in many areas that are difficult to access by humans. However, Routing remains the key challenges in sensor networks as it is important for the timely delivery of sensed datato the base station. In recent years, Multipath Routing has been used to ensure reliable and scalable data transmission in WSN. Although many multipath routing algorithms have been proposed, very few protocols have been focused on the Quality of Service (QoS) based routing. This paper proposes a QoS-aware, multipath routing protocol in which sensor nodes are clustered using the hybrid Particle Swarm Optimization-Cuckoo Search Optimization algorithm. The proposed protocol then chooses multiple stable paths (optimized network routing) using the Cluster Heads to transmit data based on multi-hop communication. Unlike the existing protocols, it relies on paths that do not affect QoS for rapid data transmission. It also extends the network lifetime by changing the Cluster Heads periodically based on the residual energy and uses the optimal number of paths to data transmission unlike the existing QoS Centric protocols. The performance of the proposed protocol is evaluated using NS-2 Simulator in different scenarios. The proposed protocol outperforms current protocols in terms of QoS parameters such as throughput, packet delivery ratio, end-to-end delay, and network lifetime, according to simulation results.

Keywords Wireless sensor network · Multipath routing · Quality of Service · Particle Swarm Optimization · Cuckoo Search Optimization

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1 Introduction

WSNs have been widely used in a variety of applications, including environmental control, disaster management, and surveillance [3, 27]. They are most often used in holistic areas where humans have difficulty in accessing. A WSN is a collection of a large number of tiny sensor nodes that are connected wirelessly for seamless communication [1]. These sensors have limited processing power, energy, memory and data transmission range. Moreover, these sensor nodes are normally transmitting data using wireless radio communication [2]. Routing is one of the key issues in the WSN because of the dynamic topological changes of the network and limited energy availability [4, 5]. It is critical when WSN is used in delay-sensitive applications such as remote patient monitoring, gas leakage detection and emergency response systems. To support such applications, it is utmost important to find the optimal route that helps to meet all Quality of Service (QoS) metrics [6]. In WSN, sensor nodes that are located close to the sink node is involved in many data transmission and transfers huge amounts of data. Subsequently, the energy levels of such nodes are drained rapidly. Hence, the network will be disconnected, which reduces the lifetime of the network. This problem is known as hot spot or energy hole problem as shown in the Fig. 1.

Many routing techniques have been proposed to address the hot spot problem in the WSN. These protocols have efficiently used the energy of nodes in the sink coverage to improve the network lifetime. Clustering is one of the efficient techniques to resolve the hot spot problem which extends the network lifetime by organizing sensor nodes into hierarchical topology. In the clustering based technique, the sensor nodes are divided into smaller groups called clusters. Each cluster has a special node called Cluster Head (CH) which is responsible for collecting data from other nodes in the cluster. CH is also responsible for data aggression and redundant data elimination. Even though CHs are used to minimize the energy depletion in the network, CHs are quickly drained due to their additional functionalities. As CHs plays an important role to determine the network lifetime, it is essential to select the best sensor nodes as CHs. Hence, CH selection is considered as an NP-Hard Problem. In recent years, many swarm intelligent algorithms that are designed based on the exploration and exploitation behavior of different birds and animals are effectively used to solve optimization problems including hot spot problems in WSN [22–26]. Furthermore, Multipath routing protocols have been recently used to improve the reliability of data transmission in WSN [7–9, 27] In this kind of routing, sensor nodes send the data over multiple paths to the sink node simultaneously. If one path fails due to the node failure or link

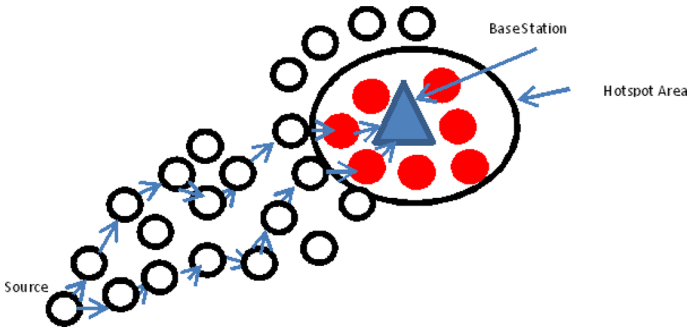


Fig. 1 The hot spot problem

failures, alternative paths are used to continue data transmission. Multipath routing is used to establish fault tolerant and reliable paths in the WSN. Multihop routing is a significant assistance required for WSN. MANET routing strategies neglect to perform well in WSN. Web steering expects to be dependable wired associations and thus there are inconsistent packet blunders, yet in WSN this isn't the situation. Additionally, MANET directing conventions accept symmetric connections between two associations which isn't the situation with WSN as the greater part of the traffic is unidirectional way.

This paper proposes a navel clustering based multipath routing protocol which uses hybrid Particle Swarm Optimization – Cuckoo Search Optimization algorithm to form clusters in WSN. It chooses the best Cluster Heads based on the residual energy and distance with the neighboring nodes to determine the optimal paths to the sink node. Perhaps, it ensures that the desired QoS metric values and energy efficiency of application never violated. It enables the source node to transmit data on multiple paths simultaneously to reduce data transmission time. Hence, it significantly conserves the residual energy of sensors and reduces the network traffic when compared to the existing protocols. Simulation results reveal that the proposed protocol improves various quality metrics such as throughput, packet loss, latency and network lifetime.

The rest of the paper is structured as follows. Section 2 discusses the principles of various multipath routing protocols and Sect. 3 presents the assumptions behind the System model. Section 4 describes the various phases of the proposed hybrid Particle Swarm Optimization-Cuckoo Search Optimization algorithm based multipath routing protocol. Section 5 accesses the performance of the proposed protocol and compares the performance with the existing protocols. Finally, Sect. 6 concludes the paper.

2 Related Work

Many routing protocols have been designed for WSNs depending on the application, architecture and objectives [1, 6, 7]. Multipath routing protocols have received more attention due to its ability to support scalability, fault tolerance and energy conservation. This section discusses the working principle of the existing multipath routing protocols.

Deepak et al. has designed a braided multipath algorithm which uses sensor nodes' information to establish two types of paths [10]. This protocol recovers from failures with less energy when compared to the earlier protocols. At the same time, it does not improve the network lifetime significantly. Swades De et al. have presented a meshed multipath routing algorithm for WSN [11]. Unlike the disjoint multipath protocol in [10], it improves the network throughput and utilizes network resources including channel bandwidth and battery power more efficiently by selective forwarding of packets and end-to-end Forward Error Correction (FEC) coding technique. However, this method needs sensor nodes with Global Positioning System(GPS).

Younis and Fahmy designed a clustering protocol called Hybrid Energy Efficient Distributed (HEED) method for multipath routing [12]. This protocol has organized sensor nodes into clusters. It then selects Cluster Head (CH) for each cluster based on the node degree and residual energy periodically. Unlike the earlier approach in [11], it did not need location-aware nodes. This protocol improves network lifetime, scalability and load balancing. In [13], Multipath Routing Protocol (MRBCH) has used credible cluster heads to ensure secure communication over multipath. In this method, CHs are selected based on the node's creditability and residual energy. Perhaps, the credit value which is calculated using trust mechanisms are

exchanged among the CHs. The CHs then formed the route to the sink node. Most of the earlier multipath protocols have tried to optimize the energy consumption for data transmission. However, many QoS based Routing protocols have been proposed to improve the reliability of critical applications while optimizing the energy consumption in the recent years.

Sohrabi et at have presented a Sequential Assignment Routing(SAR) algorithm to determine multiple paths for data transmission [14]. It organizes the nodes in tree form and selects paths to the sink based on QoS metrics, residual energy and priority level of data packets. Although SAR is energy efficient, it has to rediscover routes whenever a sensor node fails. He et al. proposed a location-aware, QoS centric protocol called SPEED [15]. This protocol ensures end-to-end delivery guarantee with desired level of QoS parameters. Felmban et al. have proposed a Multipath and Multispeed Routing Algorithm (MMSPEED) for WSN[16]. Unlike SPEED, this protocol minimizes the packet loss and network congestion by compromising QoS guarantee. Although this protocol is scalable, it does not focus on the energy consumption. The Multi-Constrained QoS Multipath Routing Protocol(MCMP) used braided multipath to transmit data over the optimal paths. It has used linear integer programming to limit the end-to-end delay in data transmission. Though it reduces the energy consumption, it does not guarantee for optimal energy consumption in all circumstances. Bagheri et al. have presented a clustering based multipath routing which uses CHs for data transmission [17]. It defines stable paths and selects cluster heads for each cluster based on the residual energy of sensor nodes in the cluster. Due to control packet overhead, it is less energy efficient and allows nodes to be fitted with a GPS system.

Mazaheri et al. suggested multipath hierarchical routing, in which sensors within a given range are clustered, and the Cluster Head is chosen from among the clusters with the highest residual energy[18].The path to the sink node is then established by within the range determined depends on the distance between Signal to Noise (SNR) ratio and buffer size. It considerably improves the reliability of paths. However, it fails to improve the latency and network lifetime. Almalkavi et al. developed another cluster based multipath protocol in which the sink initiates the cluster formation process[19]. In this algorithm, the sink node broadcasts the control packets to all sensor nodes and it selects the cluster heads based on the signal strength of node's reply. It then arranges the cluster heads into different levels and initiates data transmission from CHs in the upper level. But, it does not ensure reliability due to the existence of improper paths. SVM cluster based secure and effective routing protocol is studied in [28].

Alwan et al. developed the multipath QoSRouting (MQoSR) protocol. In this protocol, the paths are found using heuristics for neighbor selection based on geographic routing mechanism. This protocol can be used to meet the QoS requirements of various WSN applications. Deepa and Suguna have used Particle Swarm Optimization(PSO) for establishing cluster based multipath routing protocol. It uses alternate path along with primary path to transmit data. Shahbaz et al. have proposed a multipath routing algorithm which uses firefly algorithm to cluster the sensor nodes. The protocol has used fuzzy rules to establish the paths between the sources and sink node. Although it discovers two paths, it uses only one path for data transmission [9].

3 Network Model

The following assumptions about the WSN are made in the proposed work. The WSN is modeled as a weighted, undirected graph G (V,E,W). The graph model contains vertex, Edges and weight function. Here the V represents vertex set, E denotes Edge set and W represents

the weight value of communication sink. Each vertex in the Vertex Set V represents a sensor node whereas an edge in the Edge Set E represents the bidirectional communication link between the sensor nodes. A weight value is associated with each communication link. Moreover,sensor nodes are heterogeneous and have the same capabilities in terms processing power, storage, communication range and battery power. It also assumes that each sensor has a unique ID in the network.However, sensor nodes can change their transmitting range. Hence, Cluster Heads (CHs) can send data to the sink through other cluster heads and all other nodes can send data to the CH through single hop manner as shown in the Fig. 2.

With sensor node, cluster head, inter and intra cluster communication route is established in this Figure. source and sink is identified and route establishment is done properly.

3.1 Energy Model

Energy is an important in WSN as it decides the usability of sensor network. The energy depletion for transferring k bits of data over a distance d by a node i is computed as in the following Eq. 1.

$$E_T(k,d) = \begin{cases} k.E_{elec} + k\varepsilon_{fs} + d^2 & if \quad d < d_o \\ k.E_{elec} + k\varepsilon_{mp} + d^4 & if \quad d \geq d_o \end{cases} \tag{1}$$

where E_{elec} is the energy consumed by the transmitter or receiver circuitry in the sensor nodes, ε_{fs} is the energy consumed by the transmitter amplifier and ε_{mp} represents the energy consumed by the transmitter amplifier which depends on the transmission distance in a multipath model. The threshold transmission distance d_o is computed as in Eq. 2.

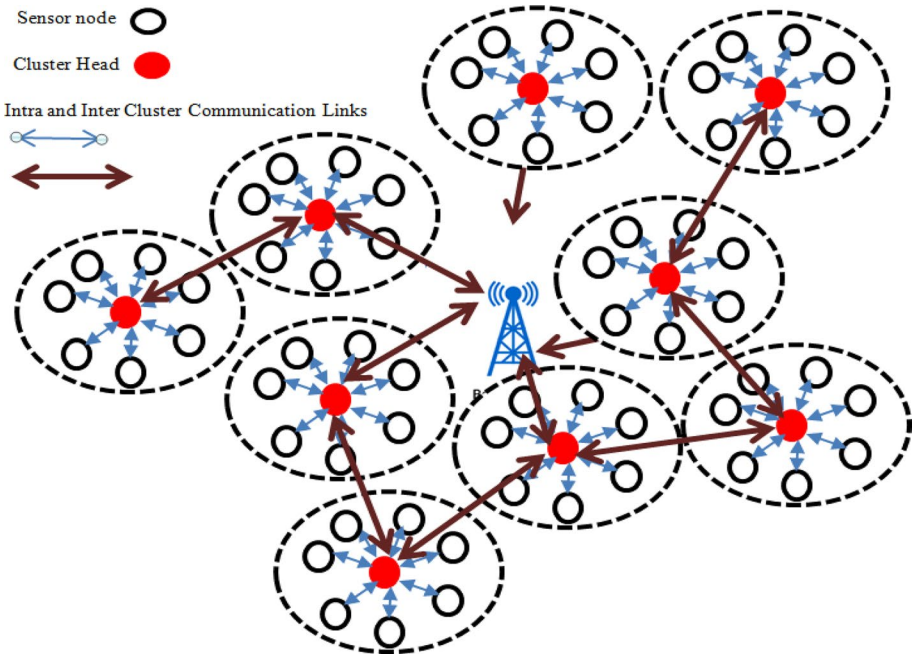


Fig. 2 System model

$$d_o = \sqrt{\varepsilon_{fs}/\varepsilon_{mp}} \tag{2}$$

Similarly, the energy required to receive k bits of data is defined as in Eq. 3.

$$E_R(k) = k \cdot E_{elec} \tag{3}$$

The residual energy is computed by subtracting the energy transmitted or received from the current energy in the sensor node.

4 The Proposed Method

The proposed routing protocol can find multiple reliable paths that support energy-efficient and satisfying the QoS requirements of the applications. It comprises the following three stages.

1. *Cluster Formation Phase* WSN is divided into many smaller clusters using the hybridization of Particle Swarm Optimization and Cuckoo Search Optimization algorithm.
2. *Route Discovery Phase* Reliable multiple paths are established and data is transmitted over multiple paths simultaneously.
3. *Route Maintenance* Rerouting or Re-clustering in case of node failures.

In the proposed method, clusters are formed using the hybrid Particle-Swarm Optimization-Cuckoo Search Optimization Algorithm in the first stage while energy efficient,optimal paths that meet the QoS requirements of application are established in the second phase[25]. The routes are maintained and reestablished in the last phase. The following subsection elaborates the processes of each phase.

4.1 Cluster Formation Phase

The sink (base station) node initiates the cluster formation process by sending Seek_Info_Msg to all nodes in its coverage area. Upon receiving the message, each node sends a reply message Source_Info_Msg containing its ID, position in two dimensional space, velocity and residual energy at that moment to the sink node. Hence, all information's about the neighboring nodes of sink node are maintained and updated in the sink node. Then, it invokes the HPSOCS algorithm for clustering the nodes. The main aspect of the hybrid PSO-CSO is that it uses the Lévy flight method to find the optimal solution in the search space [22–24, 26] as shown in Fig. 3.

Velocity of Optimization calculation is one of its significant feature, as it is the instrument used to move (advance) the situation of a molecule to look for ideal arrangements. This velocity guideline intends to accomplish a harmony among investigation and misuse. It is more efficient than the random walk method which is used in the PSO. The key steps of the HPSOCS algorithm are as follows.

Initially, a swarm with n particles that represent the sensor nodes in the sink coverage area are selected. Then the fitness value which determines the best position of the particle is calculated as in the equation.

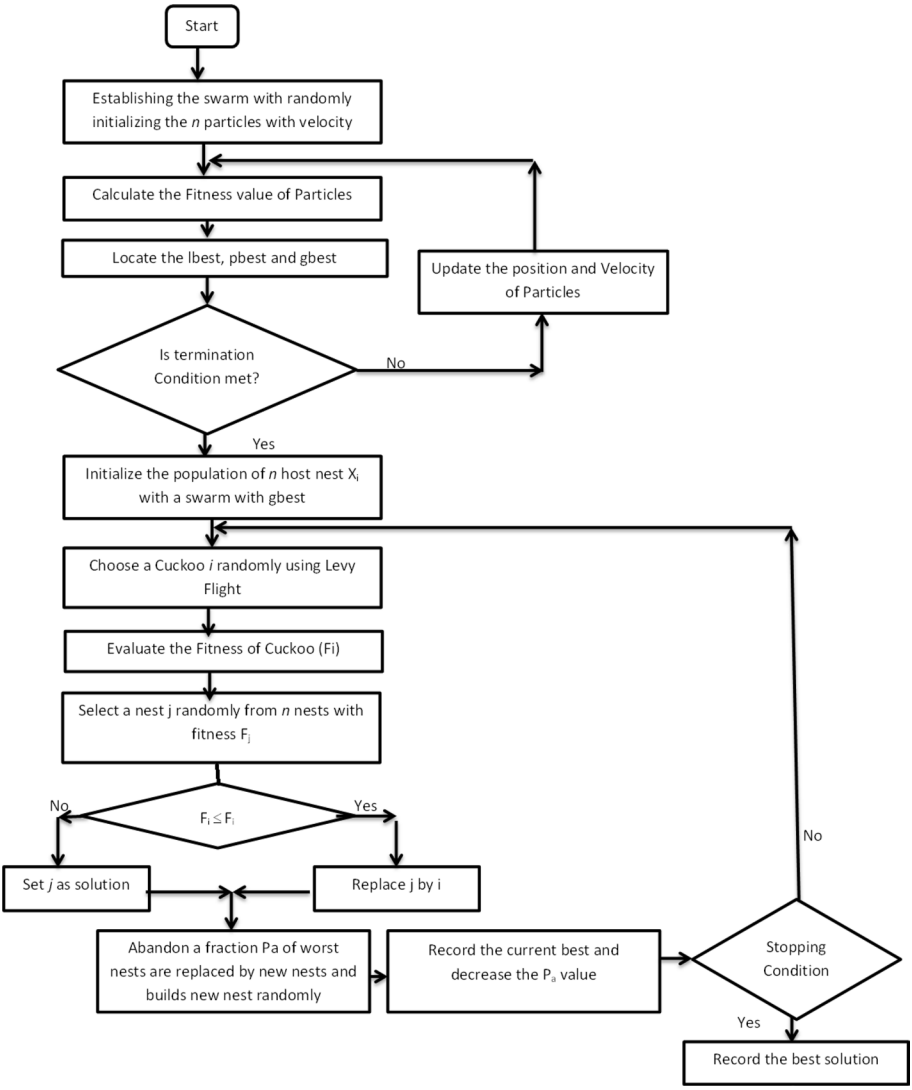


Fig. 3 Clustering using Hybrid Particle Swarm Optimization and Cuckoo Search Algorithm

*Step 1:*In the HPSOCS search space, each particle which represents the sensor node in the sink's coverage area has two dimensions, namely: particle position and velocity.

Step 2: Then the fitness value of each particle using the following equa-

$$\text{fitness(current_ particle)} = \alpha_1 \sum_{i=1}^{n} \left(distance\left(current_particle, membernode_i\right)\right)/n + \alpha_2 \sum_{i=1}^{n} (Energy\left(membernode_i\right)/Energy(current_particle) + (1-\alpha_1-\alpha_2)1/n$$

where,$\alpha_1$and α_2 are weighing parameters with values ranging from 0 to 1, and n denotes the number of sensor nodes in the cluster led by the sink node. The fitness

function is used to maximise the average distance to ensure that the best sensor node's energy is higher than the average energy of all nodes.The particle's velocity is initialized randomly. At the end of each generation, the algorithm records the best position in the current generation *pbest* and best position of the entire swarm, *gbest* and current best position *lbest* of a particle.

*Step 3.*The particles of next iteration are derived from the initial generation as follows.

Step 3.1. The velocity of the particle is updated to move towards with the best particle.It is calculated as follows:

$$\text{velocity}_{\text{new}} = \omega * \text{velocity}_{\text{old}} + r_1 c_1 (pbest - lbest) + r_2 c_2 (gbest - lbest)$$

where ω is inertia weight and the random variables r_1 and r_2 are uniformly distributed within range [0, 1]. The acceleration coefficients c_1 and $c_2$are positive constants.

Step 3.2. The new position of the particle is computed using the previous position of the particle and new velocity value.

$$\text{position}_{\text{new}} = \text{position}_{\text{old}} + \text{velocity}_{\text{new}}$$

*Step 4.*Once the particles positions are updated, the fitness values of new particles are computed as in Step 2.

*Step 5.*After that, the fitness values of the old and new particles are compared, and the better one is advanced to the next iteration.

*Step 6.*In each iteration, one best solution is selected as a*lbest* solution.The particle which has maximum fitness value up to the current iteration is selected as *gbest* solution.

// Cuckoo Search Optimization with Levy.

Step 7: Population with gbset of PSO is taken as initial population of n nests Xi where i = 1 to n.

Step 8: Fitness of each cuckoo is calculated.

Step 9: While (iteration t < Maximum Generation) or (stop criterion).

```
Select a cuckoo's egg randomlyfrom the global solution by Levy Flights
        Xi(t+1)=Xi(t) + α*L ,
where L is a value from the Levy distribution and α is the step- size

Determine its fitness level.
Choose a nest among n (say j) at random
if (Fitnessi>Fitnessj )
        Replace the nest with cuckoo
end
A fraction (pa) of the worse nest is abandoned and new nests are added
Compute thefitness of new nests and maintain the best nests
Rank the nestswith the optimal value
The best nest is passed to the next generation

End while
End
```

*Step 10.*Once the algorithm terminates, the best nest with the maximum fittest cuckoo egg is considered as the optimal solution and Cluster Heads (CHs) are generated from the optimal solution.

Lévy Flight, which depends on the flight conduct of creatures and bugs, has been differently concentrated in literary works. It is characterized as a sort of non-Gauss stochastic cycle with an arbitrary walk whose progression sizes depend on Lévy stable dispersion. There are bunches of renditions of Lévy dispersion. Mantegna Algorithm has been tried to be the most proficient technique to produce Lévy circulation esteems. It has been received in numerous development calculations to escape from neighborhood least. The CHs are then broadcast thejoin_Cluster message containing their location to other sensor nodes in its communication range. When a sensor node receives the message, it sends member_Cluster to the CH. Upon receiving reply messages to all its request message, CH includes the nodes in its cluster and sends ack_CH message to the node. When the sensor node receives multiple join_Cluster messages, it sends member_Cluster message to the CH which is nearest in terms of Euclidean distance and sends reject_Cluster messages to other CHs.

4.2 Route discovery

In this step, the routes between the CHs and the path between the CHs and the sink node are established. At first, each CH calculates the score using three factors, namely Residual Energy, Distance and traffic support. The traffic support of CH is calculated as the ratio between thenumber of packets in the buffer to the total capacity of the CH buffer. After that, CH broadcasts a message containing its score to the nearby CHs which is subsequently stored in the Nbr-CH table in the CH. This process is repeated until all the CHs have complete information about all neighboring CHs.When a CH receives a request to transfer data to the base station, it sends its application specific requirements of QoS parameters such as Reliability, Delay and Energy to the CH through route_ Request message. Upon receiving the message, CH looks for appropriate CHs in the Nbr-CH table and forwards the request to the CHs that can fulfill the requirements. This process continues until the CH establishes the multiple paths that satisfy the QoS requirements of the sensor nodes to the sink node as shown in Fig. 4. When there is a path failure and CH replacement due to low residual energy, the route discovery process is initiated again. When the base station receives the route_ Request message, it sends the route_ Reply message through the established path. Each CH waits for the route_ Request message from their neighboring CH for a specific time T_{wait}. It uses all the paths established within the time frame to transmit the data simultaneously.

Once the paths are established, all paths are used by the source node to deliver packets about events to the sink nodes. The data transmission algorithm is shown below:

```
procedure Data_Transmission
begin
        while ( there is a data about a event)
        begin
                if( number of path == 1) then
                        Send the date through the single path
                else

                        Send the data over the multiple paths in round robin manner
                        Records the end-to-end energy consumption of the path
                        Sort the paths based on end-to-end energy consumption
                end

  end
```

The proposed protocol uses a path for a while and selects another path for the next time period when there is more than one path to the sink. One of the notable improvements in the proposed protocol is that it utilizes all paths established to deliver packets. This is because of utilizing the energy- efficient paths for some time longer than the other paths. Since it changes the paths and uses them orderly, it ensures the reliability of the paths and network lifetime.

4.3 Route Maintenance

The sink node of the network initiates the rerouting process when the QoS parameter such as throughput drops below the threshold required by the application. Moreover, the cluster heads maintain the residual energy of all sensor nodes within the cluster and the sink node maintains the residual energy of all cluster heads. Initially, the sink

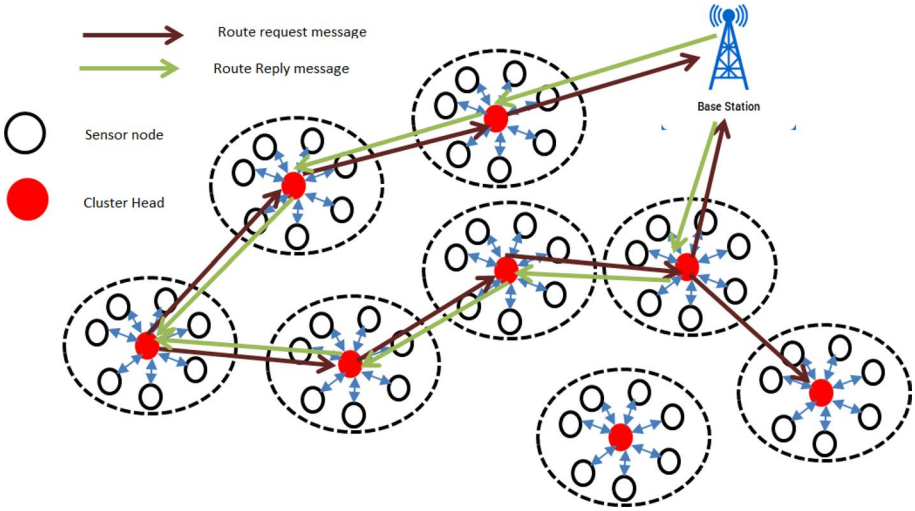


Fig. 4 Route establishment process

node initiates the re-clustering process when the residual energy of the cluster head falls below the average residual energy of CHs which is computed initially. In this manner, the proposed protocol will increase the network lifetime and utilize the energy in the sensor nodes more efficiently.

5 Performance Evaluation

The proposed method is implemented using NS2.35 simulator and its performance is compared with the existing protocols based on EE-LEACH, EPSO-CEO, OQoS-CMRP. The simulation parameters are shown in Table.1. The sensor nodes are placed randomly on the 300 m × 200 m and the number of nodes is varied from 50 to 250 during the simulation. IEEE 802.15.3 is used to define the physical and MAC sub layer for data packet transmission. The size of data packets is 128 bites and the initial energy of each sensor node is set at 3 J. The rectangular field is used to represent the network coverage area and the network has supported bandwidth of 250 kbps.

Table 1 Simulation parameters

Radio propagation model	Two-ray ground reflection model
MAC Type	IEEE 802.15.4
Antenna Type	Omnidirectional
Simulation area	300×200 m
Link bandwidth/data rate	250 kbps
Radio frequency	2.4 GHz
Number of nodes	50, 100, 150, 200, 250
Radius (r)	40 m
Simulation time	500 s
Channel type	Channel/wireless
Queue type	Priority queue
Data packet size	128 bytes
Control packet size	32 bytes
Initial energy	3 J
Transmission power consumption	0.002*dist J
Reception power consumption	0.02 J
E_{elec}	50 nJ/bit
ε_{fs}	1011 J/bit/m2
ε_{mp}	1.3 1015 J/bit/m4
d_o	87 m
Acceleration coefficient C_1,C_2	2
Maximum iteration of PSO	200
ω	0.5
P_a	0.25
Levy component L	1.5
Maximum iteration	300
Traffic source	CBR

The performance of the algorithm is evaluated using various QoS parametes such as throughput, Packet Delivery Ratio, end-to-end delay, network lifetime and energy consumption as defined in [20].

5.1 Network Life Time

The network's lifespan is determined by the number of active sensor nodes and the WSN's route sustainability. The sensor node will be active as long as its energy is not drained fully. The proposed protocol is compared with the existing protocols in terms of network lifetime. As shown in the Fig. 5, the network life time increases with the number of nodes in the network irrespective of protocols. However, the proposed protocol has significantly improved the lifetime when the number of nodes compared to the EE-LEACH,EPSO-CEO and OQoS-CMRP protocols due to the selection and replacement of cluster heads.

5.2 Energy Consumption

The energy consumption of node is computed using the following equation. The total energy consumption is computed as a sum of all nodes energy consumption in the network.

$$E_{consumption} = E_{initial} - E_{residual}$$

The energy consumption is steadily increased as the number of nodes is increased in the network. As shown in Fig. 6, the energy consumption of proposed algorithm is decreased by 40.06%, 32.4%, and18.21% compared with EE-LEACH, EPSO-CEO and OQoS-CMRP protocols.

5.3 Throughput

It is defined as the number of data packets received by the sink node from the source node at a particular time period. It is found that all four protocols have almost the same throughput when the simulation time is less than 200 s as in Fig. 7. However, the proposed protocol significantly improves the throughput as the simulation time increases when compared to the other three protocols when the simulation time is increased.

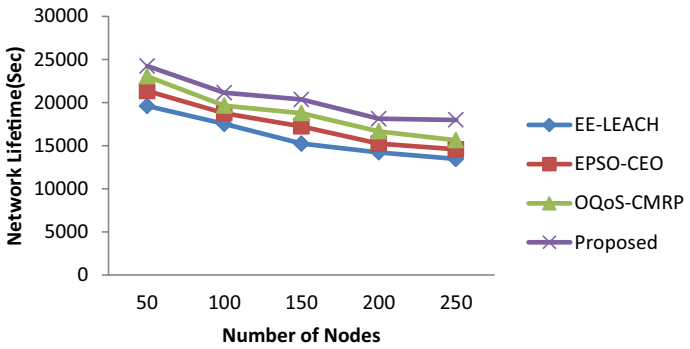


Fig. 5 Comparison of network lifetime

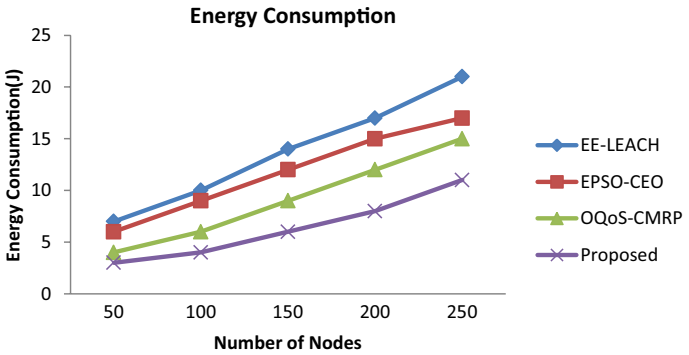


Fig. 6 Comparison of energy consumption

5.4 Packet Delivery Ratio (PDR)

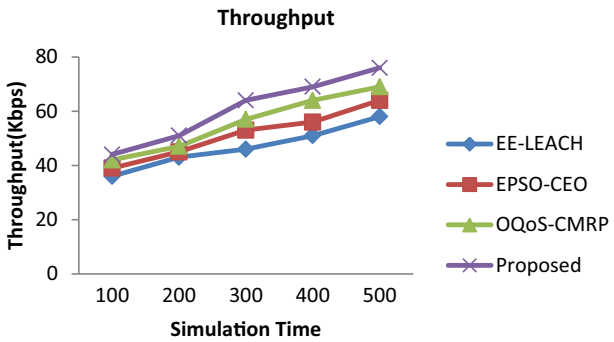
It is the ratio between the numbers of data packets delivered to the sink node and number of packets sent by the source node. It is found that the delivery ratio of protocol is more than 5.55% higher than the OQoS-CMRP protocol as shown in Fig. 8.

5.5 End-to-End delay

That's the cumulative time it takes to send a packet from a source node to a sink node.The proposed protocol has optimal end to end delay as compared to the the EE-LEACH, EPSO-CEO and OQoS-CMRP protocols.

As shown in the Fig. 9 the EE-LEACH protocol has maximum end to end delay as compared to other three routing protocols. But, the proposed protocol considerably reduces the end to end delay as the simulation time is increased when compared to the OQoS-CMRP protocol. This is because of selection routes with less traffic during route establishment phase. QoS adopted performance of proposed method achieves the result by comparing above results. Depends on the performance of delay, PDR, throughput and network lifetime, the QoS is analyzed. Three different models are compared to proposed work, which obtains the better outcome.

Fig. 7 Comparison of throughput



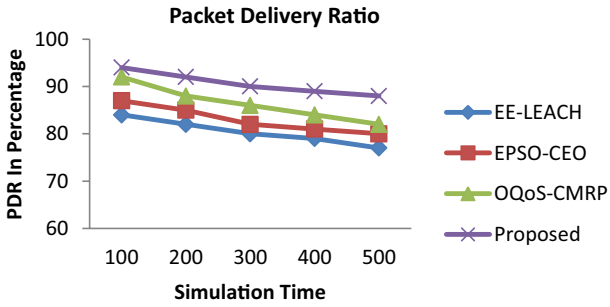


Fig. 8 Comparison of packet delivery ratio

6 Conclusion

The proposed QoS-aware Multipath Routing (QMR) protocol has established multiple optimal paths between the source and sink node in the wireless sensor network (WSN). In the protocol, the hybrid Cuckoo Search-Particle Swarm Optimization algorithm is used for clustering the sensor nodes and chooses the cluster heads to ensure the reliable data delivery. The proposed protocol uses several paths for delivering data packets and has excellent control over data traffic in the network. Simulation results show that the proposed protocol outperforms the existing QoS centric protocols like EE-LEACH, EPSO-CEO and OQoS-CMRP in terms of throughput, end-to-end delay, Packet Delivery Ratio, network lifetime and energy conservation. Hence, the protocol can be used in various delay sensitive applications. In future, the efforts should be made to improve the security aspects of the protocol. Moreover, the performance of the protocol will be assessed in Wireless multimedia sensor networks. In future, the work algorithm is improved with other deep learning and swarm intelligent techniques.

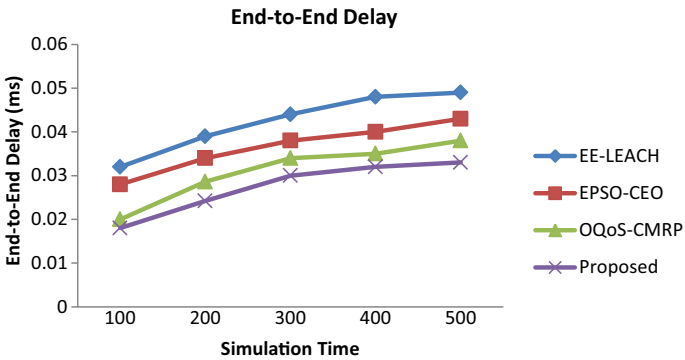


Fig. 9 Comparison of end to end delay

Declarations

Conflict of interest Conflict of interest is not applicable in this work.

Ethics Approval and Consent to Participate No participation of humans takes place in this implementation process.

Human and Animal Rights This article does not contain any studies with human participants or animals performed by any of the authors. No violation of Human and animal rights is involved.

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